The Role of Inflammation in Alzheimer’s Disease

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Study Aims

Aim 1:

- ***H1a***: Higher baseline peripheral inflammatory marker levels will be associated with greater 1-year decline in cognitive outcomes and AD-signature mean cortical thickness

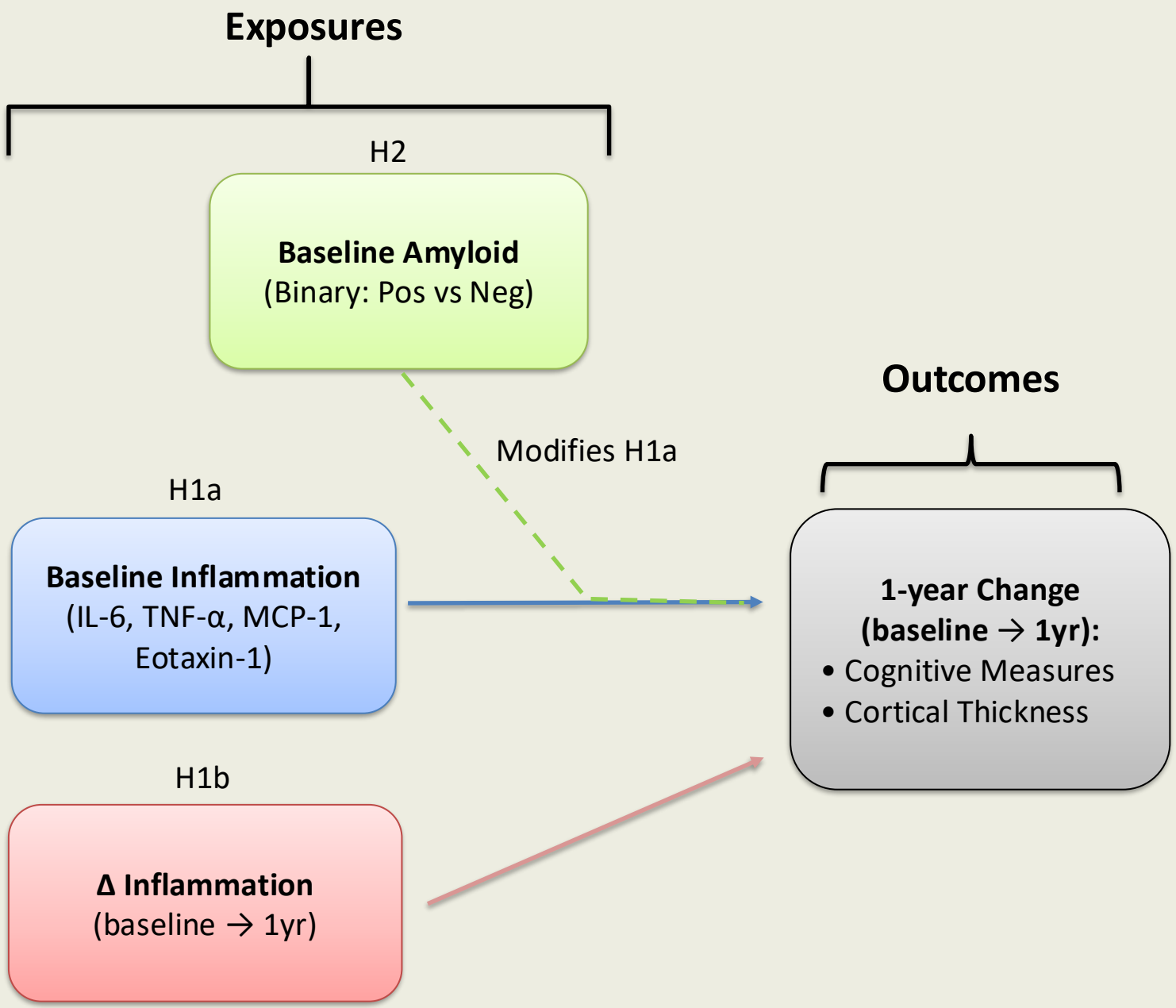
- ***H1b***: Greater 1-year increases in peripheral inflammatory markers will be associated with greater 1-year declines in the same cognitive and cortical outcomes.

Aim 2:

- ***H2***: Amyloid deposition will modify the association between inflammation markers and 1-year decline in cognitive and AD-signature cortical thickness.

Analysis

- ❑ Linear regression models
- ❑ Adjust for baseline outcomes + baseline covariates
  - ➢ Sensitivity for covariates
    - APOE, CDR, etc
- ❑ Multiplicity
  - ➢ α=0.05/4 independent tests
  - ➢ Outcomes (2): Cognitive & Cortical
  - ➢ Predictors (2): Cytokines & Chemokines



Power – Aim 1

- Estimated using correlation-based methods
- Evaluated across multiple α levels
  - Minimal impact on choice of α
- 80% power at |r| ≈ 0.25–0.29

Aim 1 Power at N = 175

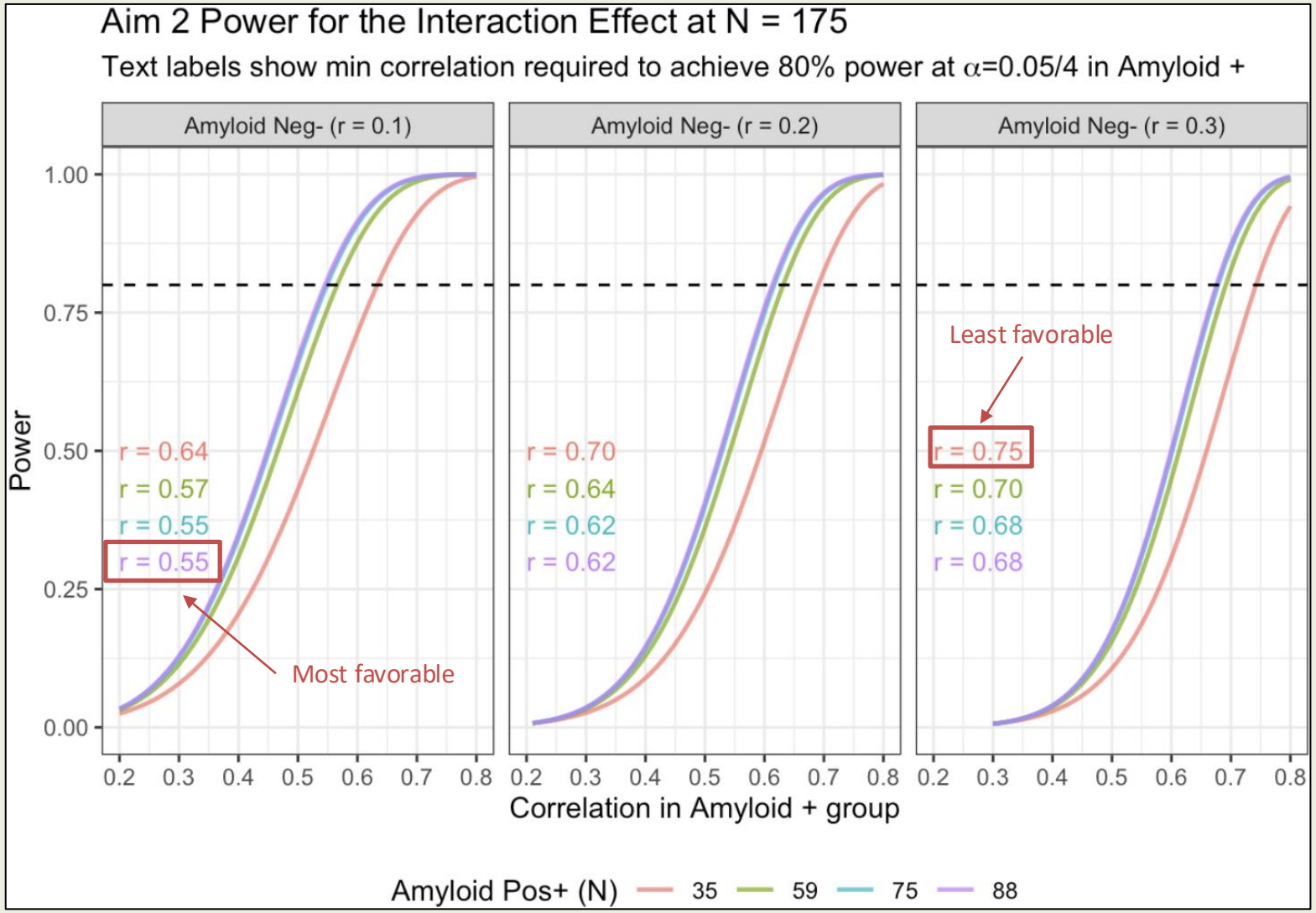
Power (%) across assumed correlations and significance thresholds

	Power (%)		
Assumed |r|	**α = 0.05/4**	**α = 0.05/10**	**α = 0.05/20**
0.50	100.0	100.0	100.0
0.45	100.0	100.0	100.0
0.40	99.9	99.7	99.5
0.35	99.0	97.7	96.3
0.30	94.2	89.7	85.3
0.25	80.6	71.0	63.2
0.20	56.7	44.4	36.1
Min |r| for 80% power	**25.0**	**28.0**	**29.0**

Minimum detectable correlation corresponds to the smallest absolute correlation achieving at least 80% power.

Power – Aim 2

- Estimated using two-sample correlation method
  - α=0.05/4 = 0.0125
    - Per Aim 1
- Power driven more by difference between groups
- 80% power requires:
  |r|≈ 0.55–0.75 in Amyloid+ group



Conclusions

- **Power:**
  - Aim 1: Moderate effects detectable
  - Aim 2: Moderate-to-large interactions detectable

- **Limitations:**
  - Limited power for small effects
  - Interaction effects harder to detect
  - Sample size is likely too small